Company Stability Prediction

Machine Learning – Classification Use Case

1. Objective

Predict whether an employee is likely to stay with the company or leave, based on the employee feedback collected about the company.

2. Assumption

Feedback is available from each employee about the company. The feedback is represented as ratings across multiple questions.

3. Input and Expected Output

Input	Expected output
Employee feedback (ratings for Q1, Q2, Q3, Q4, etc.)	Employee retention prediction: Will Stay / Might Not Stay

4. Proposed Machine Learning Stages

Stage 1 – Data preparation: Collect and organize employee feedback; convert the responses into usable rating data.

Stage 2 – Supervised learning: Use previously labelled examples (employees who stayed or left) to train the model.

Stage 3 – Classification: Classify an employee into a retention outcome, such as likely to stay or likely to leave.

5. Sample (Dummy) Dataset

The following sample is transcribed from the handwritten notes. The third employee's first rating is obscured in the photograph, so it is shown as "—".

Employee	Q1	Q2	Q3	Q4	Aggregate rating
Employee 1	1	2	3	2	To be calculated
Employee 2	5	4	5	4	To be calculated
Employee 3*	—	3	3	3	To be calculated
Employee N	…	…	…	…	To be calculated

*Employee 3 is a best-effort row label; the original row label and Q1 value are partly covered.

6. Illustrative Decision Logic

Calculate the aggregate rating (for example, the average of Q1–Q4) for each employee. The handwritten notes appear to suggest the following interpretation:

Average rating	Predicted outcome
Below 3	Will leave
Above 3	Will stay
Equal to 3	Might leave / uncertain

Note: The threshold interpretation above is inferred from the handwriting (which explicitly shows "average < 3 → will leave" and "might leave"). Please confirm the exact rule for an average of 3 and whether the "will stay" condition is average > 3 before using this as a model specification.

7. Overall Workflow

Employee feedback → Rating data preparation → Aggregate rating calculation → Supervised model training → Classification → Retention prediction